

Ausbildungsinhalte

Die Teilnehmer*innen haben in 720 Unterrichtseinheiten folgende Inhalte gelernt, diskutiert und in unterschiedlichen Aufgaben und Projekten vertieft:

Programming & Database

Python

- Pandas
- Scikit-Learn
- Statsmodels
- TensorFlow/Keras

Visualization

- Matplotlib
- Seaborn
- Plotly
- Altair

Unix

SQL

Git (Github)

Exploratory Data Analysis (EDA) & Statistics

Data Analysis

- Distributional Analysis
- General Statistics

Data Visualization

- Distribution Plots
- Relationship Plots
- Geo Visualizations

Data Science Portfolio Projects

Project 1

- Exploratory Data Analysis

Project 2

- Predictive Modeling

Final Project

- Capstone (4 weeks)

Machine Learning Algorithms

Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forests
- Naive Bayes
- KNN
- SVM

Unsupervised Learning

Deep Learning

Natural Language Processing

Time Series

Evaluation & Performance metrics

Confusion Matrix

Regression and Classification

Metrics

Error Analysis

Model tuning

- Bias-Variance Tradeoff
- Regularization
- Cross-validation

Optimization

- Gradient Descent
- Cost Functions

Collaborative Working & Social Learning

Pair Coding

- Driver & Navigator

Agile Workflow

- Daily Stand-Ups
- Kanban Project Board

Daily Class Review

- Team Work & Self-organization Skills

Group Work, Individual

Exercises, Reversed

Classroom (Presentations from Students)

Career Coaching & Mentoring

Machine Learning Engineering

MLFlow Tracking

Serving Machine Learning

Models

Online Dashboards

Cloud (Intro to GCP)

Communication w/Stakeholders

Business presentation

- Non-technical Stakeholders

Data Science Lifecycle presentation

- Technical Stakeholders

Stakeholder Review

- Compact & Time-critical Project Alignment